# 🛠️ 02: TECHNICAL ARCHITECTURE & SECURITY REPORT

## DigyNex 360 Ecosystem (2026 Edition)

### 🏗️ 1. SYSTEM TOPOLOGY

The DigyNex 360 ecosystem is built on a **Modular Micro-Services Architecture**, ensuring horizontal scalability and fault tolerance.

- **Intelligence Hub Landing:** High-fidelity HTML/CSS gateway featuring the **Lead Capture Portal** (Executive Access Gate).
- **Frontend Engine:** Vue 3 (Composition API) with Tailwind CSS for high-fidelity, reactive UI/UX.
- **Database Backbone:** Supabase (PostgreSQL) providing real-time synchronization and secure data storage.
- **Automation Engine (n8n):** Enterprise-grade n8n cluster for ETL pipelines and AI-augmented workflows.
  - **Strategic Intelligence Node:** Connects live database metrics to the Gemini LLM for predictive ROI modeling and pushes real-time updates to the executive dashboard.
- **AI Models:** Deep integration with Google Gemini 1.5 Pro and Imagen 4.0 for generative intelligence & photorealistic asset generation.

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### 🛡️ 2. SECURITY & SOVEREIGNTY (Zero-Trust)

Security is at the foundation of the DigyNex architecture.

- **Siloed Multi-Tenancy:** We utilize a `<tenant_id>` based isolation layer. Data bleeding is mathematically impossible within our PostgreSQL schema.
- **Auth Shielding:** JWT-based authentication via Supabase Auth, securing every RPC and API call.
- **Docker Containerization:** Applications are deployed in isolated capsules, allowing for seamless **On-Premise Deployment** for high-security corporate clients.
- **Lead Capture Isolation:** All "Executive Access" requests are indexed with PII encryption before being routed to the CRM.

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### ⚡ 3. THE "PROJECT NEXUS" ETL PIPELINE

Project Nexus uses an asynchronous processing model to handle high-volume data ingestion.

- **Parser Engine:** Automatically identifies schemas from CSV and PDF manifests.
- **Validation Layer:** Cross-references incoming data against Master Ledgers to prevent cost-variance drift.
- **Financial Integrity:** Immutable transaction logs for every bulk-import event.

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### 📊 4. SCALING CAPABILITIES

The system is designed to scale dynamically from **500 to 1,000,000+ active profiles** without database latency, optimized for global corporate growth.

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*Technical Authority: DigyNex Engineering Division*